



New Course Proposal: AERO 100 - BASIC UAV OPERATOR

Co-Contributor

Please visit the Cuesta CurriQunet Meta Help Page for help, instructions, and tutorials:

https://www.cuesta.edu/meta/docs/Help_Documents.pdf

Contributor

Open the Form Properties to select co-contributors and assign permissions.

Basic Course Information

Overview

Is this a common course? No

Banner Course Number

Discipline AERO

Course Number 100

Full Course Title BASIC UAV OPERATOR

Abbreviate Title Basic UAV

Proposal Rationale

This course is designed to meet the need for remote unmanned aerial vehicle operators in the district. The FAA requires pilots utilizing small unmanned aerial vehicles to have an FAA Airman Certificate pursuant 14 CFR Part 107. Unmanned Aerial Vehicle technology is a growing industry as San Luis Obispo County is emerging as a leader in the design and manufacture of these aircraft. As this technology gains acceptance, there is anticipation of increased usage in existing college programs to include survey/ civil engineering, real-estate and construction, criminal justice, agriculture, transportation and logistics, fire technology, as well as aviation maintenance programs. This course is a foundational course as part of a broader integration of remote applications

Catalog Description

Prepares students to meet the Federal Aviation Administration requirements for certification to operate Unmanned Aircraft Vehicles under 55 pounds for commercial operations pursuant to 14 CFR 107. Emphasizes the safe, legal, and ethical aspects of UAV operations in civil, public, and commercial settings. Includes special emphasis on capabilities and limitations of optical payloads and sensors, flight operation planning, and safety. Prepares students to safely operate UAV through all phases of flight to National Institute of Standards and Technology Level 2 Open Area Maneuvering remote pilot proficiency standards.

Schedule Description

Prepares students to meet the Federal Aviation Administration requirements for certification to operate Unmanned Aircraft Vehicles under 55 pounds for commercial operations pursuant to 14 CFR 107. Emphasizes the safe, legal, and ethical aspects of UAV operations in civil, public, and commercial settings. Includes special emphasis on capabilities and limitations of optical payloads and sensors, flight operation planning, and safety. Prepares students to safely operate UAV through all phases of flight to National Institute of Standards and Technology Level 2 Open Area Maneuvering remote pilot proficiency standards.

Implications**Budget No****Justification****Facilities Yes****Justification**

Flight operations may be conducted outside the normal classroom, include open areas where flight operations can be safely conducted.

Equipment Yes**Justification**

Will require small UAV's for practical application.

Staffing Yes**Justification**

14 CFR Part 107 Drone Operators are needed to safely oversee student flight operations.

Similar Course No**Justification****Assignments ARE College Level Yes****ESL/Basic Skills Course No****Distance Education No****Correspondence Education No****Component of Program****Credits and Hours**

Course Type D - Credit - Degree Applicable**Course has variable hours No****Hours****Min Lecture Hours - in class 2.000****Min Lab Hours - in class 3.000****Min Activity Hours - in class****Min Work Experience****TBA Hours per term****Min Lecture Hours - out of class****Min Lab Hours - out of class****Min Additional Instructor Contact Hours Per Week****Min Work Experience****Max Hours****Max Lecture Hours - in class****Max Lab Hours - in class****Max Activity Hours - in class****Max Work Experience****Max Lecture Hours - out of class****Max Lab Hours - out of class****Max Additional Instructor Contact Hours Per Week****Max Work Experience**

Total Student Hours and Credit

	Hours/Week	Hours/Term
Lecture Hours - in class	2.00	36.00
Lecture Hours - out of class	4.00	72.00
Lab Hours - in class	3.00	54.00
Lab Hours - out of class	0.00	0.00
Activity Hours - in class	0.00	0.00
Activity Hours - out of class	0.00	0.00
Work Experience Hours	0.00	0.00
TBA Hours per term		0.00
Total Student Hours per term		162.00
Hours per unit Divisor		54
Units of Credit		3.00

Grading Method

- Letter Grade Only

Repeatability

For repeatability information, please consult the college catalog

Is this course repeatable? No

Number of times this course can be repeated for credit

Justification of Repeatability

Cross-Listed Courses

This Course Is Cross-Listed No

This course is part of the following cross listing

Minimum Qualifications

Minimum Qualifications

1. Minimum Qualification Aeronautics

Condition	
Comments	

Methods of Instruction

Lab Yes

Lecture Yes

Activity No

Distance Education Yes

Correspondence Education No

Work Experience No

Directed Study No

Field Trips No
 Field Experience No
 Tutor Non-credit No
 DE Synchronous Yes
 DE Asynchronous Yes
 Lecture/Lab Yes

Student Learning Outcomes

If this course is not part of a program of study, click [here](#) to map these SLOs directly to ILOs. Yes

Outcome

1. **Outcome Text**

Demonstrate mastery of FAA Part 107 small unmanned aerial vehicle knowledge topics, including airspace, aviation weather, aircraft loading, emergency procedures, crew resource management, aeronautical decision-making, and airport operations, by passing a written exam on these topics with a minimum passing score of 80 percent.

Institution Outcome Map

- **CLO(ILO) to SLO Map Top ILO Grouping(Delta)**
 - a. Recognize, assess, and demonstrate the skills and behaviors that promote academic and professional development
- **CLO(ILO) to SLO Map Top ILO Grouping(Delta)**
 - a. Draw conclusions based on the scientific method, computations or experimental and observational evidence.

2. **Outcome Text**

Safely maneuver a small unmanned aerial vehicle through an open test lane using the National Institute of Standards and Technology Level 2 standard course within the NIST-prescribed time limit (25 minutes).

Institution Outcome Map

- **CLO(ILO) to SLO Map Top ILO Grouping(Delta)**
 - a. Recognize, assess, and demonstrate the skills and behaviors that promote academic and professional development
- **CLO(ILO) to SLO Map Top ILO Grouping(Delta)**
 - c. Demonstrate the professional skills necessary for successful employment

Objectives

Objectives

If an Objective cannot be deleted, make sure a Content-Review found in the Content Validation Page is not using that objective.

1. **Objective**

Perform preflight inspection to identify maintenance deficiencies that adversely affect aircraft performance.

Methods of Evaluation

- Class Performance(s)
- Quizzes/Exams
- Performance Exams

Other No

Other Methods

2. Objective

Calculate the performance of UAV to include weight and balance, takeoff and climb, endurance, and aerodynamic effects based on actual and forecast aviation weather.

Methods of Evaluation

- Class Performance(s)
- Quizzes/Exams
- Performance Exams

Other No

Other Methods

3. Objective

Implement appropriate decision making and judgment to mitigate risk based on human factor considerations.

Methods of Evaluation

- Field Work
- Performance Exams

Other No

Other Methods

4. Objective

Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.

Methods of Evaluation

- Quizzes/Exams
- Essay Exams
- Performance Exams

Other No

Other Methods

5. Objective

Perform a flight of an unmanned aerial vehicle to the "Open Lane Level II NIST Standard Test Methods for small Unmanned Aircraft Systems (sUAS). The U.S. National Institute of Standards and Technology (NIST) quantitatively evaluates various system capabilities and remote pilot proficiency.

Methods of Evaluation

- Performance Exams

Other No

Other Methods

Topics and Scope

Detailed Topical Outline

Add topics individually, one at a time.

1. Topic

Maintenance and Preflight Inspection Procedures

- Regulatory Requirements for Maintenance
- Preflight Inspection
- Postflight Inspection
- Manufacturer Inspection Intervals
- Battery Maintenance
- Flight and Maintenance Logs
- Maintenance, Repair, and Replacement of Components

Objectives

- Perform preflight inspection to identify maintenance deficiencies that adversely affect aircraft performance.
- Perform a flight of an unmanned aerial vehicle to the "Open Lane Level II NIST Standard Test Methods for small Unmanned Aircraft Systems (sUAS). The U.S. National Institute of Standards and Technology (NIST) quantitatively evaluates various system capabilities and remote pilot proficiency.

2. Topic

UAV Flight Operations

- Takeoff
- Aircraft Control
- Preflight Programming
- Turns around a point
- Aircraft Positioning
- Altitude Control
- Sensor Zoom/Pan/Tilt
- Airspeed Control
- Approach
- Landing

Objectives

- Perform a flight of an unmanned aerial vehicle to the "Open Lane Level II NIST Standard Test Methods for small Unmanned Aircraft Systems (sUAS). The U.S. National Institute of Standards and Technology (NIST) quantitatively evaluates various system capabilities and remote pilot proficiency.

3. Topic

Emergency Procedures

- Contingency Planning
- Ground Emergencies
- Inflight Emergencies
- Emergency Communications

Objectives

- Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.

4. Topic

Human Factors

- Health and Environmental Factors
- Decision Making and Judgment
- Risk Management and Aeronautical Decision Making

Objectives

- Implement appropriate decision making and judgment to mitigate risk based on human factor considerations.

5. Topic

Aircraft Performance

- Weight and Balance Calculations
- Density Altitude
- Takeoff and Climb Performance
- Endurance
- Landing Distance
- Aerodynamics
- Ground Effect

Objectives

- Calculate the performance of UAV to include weight and balance, takeoff and climb, endurance, and aerodynamic effects based on actual and forecast aviation weather.

6. Topic

Aviation Weather Theory, Sources, and Effects on Performance

- Fronts, Stability, Clouds, and Thunderstorms
- Fog
- Structural Icing and Frost
- Weather Sources for Aviation Operations
- The AWC Home Page (NWS Aviation Weather Center)
- Practical Sources for the Remote Pilot
- Automated Weather Reporting Stations

Objectives

- Calculate the performance of UAV to include weight and balance, takeoff and climb, endurance, and aerodynamic effects based on actual and forecast aviation weather.
- Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.

7. Topic

Airport and Off-Airport Operations

- Operations at Non-Tower Airports
- Operations at Airports with Control Towers
- Off-Airport Operations
- Flying at Night

Objectives

- Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.

8. Topic

Airspace and Navigation

- Aeronautical Charts
- Knowing Where You Can Fly
- Geographical Coordinates
- Chart Reading
- Airspace
- Correcting for Wind Drift
- Special Use Airspace

Objectives

- Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.

9. Topic

Regulations, Procedures, and Other Aeronautical References

- UAS Terms and Abbreviations
- Types of UAV, Components and Uses of UAV
- Government Regulations and Other Procedures Publications
- FAA Chart Supplements
- Advisory Circulars
- Notices to Airmen
- Aviation Media

Objectives

- Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.

Assignments

Add assignments individually, one at a time. Include a minimum of two Critical Thinking examples, and a minimum of two Independent Work examples.

1. Assignment Type

Assignment

Perform preflight calculations to determine if flight can be safely conducted based on aircraft performance, knowledge of aviation weather, forecast products, and airspace.

Course Objective

- Calculate the performance of UAV to include weight and balance, takeoff and climb, endurance, and aerodynamic effects based on actual and forecast aviation weather.
- Implement appropriate decision making and judgment to mitigate risk based on human factor considerations.
- Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.

2. Assignment Type

Assignment

Perform small unmanned aerial vehicle flight to the standard test methods found in NIST sUAS open lane Level II standards by maneuvering a sUAS over selected targets in specific sequence and land aircraft safely.

Course Objective

- Perform a flight of an unmanned aerial vehicle to the "Open Lane Level II NIST Standard Test Methods for small Unmanned Aircraft Systems (sUAS). The U.S. National Institute of Standards and Technology (NIST) quantitatively evaluates various system capabilities and remote pilot proficiency.

3. Assignment Type

Assignment

Based on knowledge of the weather, approved aviation weather forecast sources, and mission requirements, determine if flight can be safely conducted near an airport with an operating control tower.

Course Objective

- Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.

4. Assignment Type

Assignment

Perform a technical preflight inspection of aircraft in accordance with the manufacturer's recommendations, determine if safe for flight, and calculate expected in-flight endurance.

Course Objective

- Perform preflight inspection to identify maintenance deficiencies that adversely affect aircraft performance.
- Calculate the performance of UAV to include weight and balance, takeoff and climb, endurance, and aerodynamic effects based on actual and forecast aviation weather.
- Implement appropriate decision making and judgment to mitigate risk based on human factor considerations.

Methods of Evaluation

Evaluation Method

Please select all items listed except "Other" (select "Other" only if needed).

- Field Work
- Class Performance(s)
- Quizzes/Exams
- Essay Exams
- Performance Exams

Other No

Other Methods

Course Cap

Course Cap 25

Face to face 25

DE modality 25

Course Category 9 - Lab - Small (20-30)

Chosen Cap is within the selected range of the definition

Assignments

Select at least one assignment that aligns with the course category selected.

- Perform a technical preflight inspection of aircraft in accordance with the manufacturer's recommendations, determine if safe for flight, and calculate expected in-flight endurance.

AssignmentType

- Perform small unmanned aerial vehicle flight to the standard test methods found in NIST sUAS open lane Level II standards by maneuvering a sUAS over selected targets in specific sequence and land aircraft safely.

AssignmentType**Methods of Evaluation**

Select at least one method of evaluation that aligns with the course category selected.

- Performance Exams
- Field Work

- Class Performance(s)

If you have additional alignment in objectives or another area in your COR then explain this in the following box

Library

Typical library usages for students in this course are

Access to computers for obtaining real-time weather for preflight calculations and internet access for aircraft performance capabilities and limitations.

Library currently has necessary resources for students to complete course assignments. Yes

Text, Reading, and Materials

Textbook

1. **Author** David Ison
Title The Complete Remote Pilot
Edition 3rd
City Newcastle, Washington
Publisher Aviation Supplies & Academies, Inc.
Year of Publication 2025
Rationale for Textbooks Older than 5 Years

OER

Manual

1. **Author** DJI
Title Mavic Drone Manua
Pub Date 2025
Publisher DJI

Periodical

Software

Other

1. **Description**
Access to online aviation weather services provided by the National Weather Service - Aviation Weather Center.

Requisites & Advisories

This course has requisites. No

Requisites

Any requisites which are used in the Content Review section below must be removed before removing the requisite in this section.

Content Review

For all course prerequisites, corequisites, and advisories, you will need to demonstrate that the content of the proposed course as reflected in the course outline clearly builds upon and requires the content of the prerequisite, corequisite, or advisory. To demonstrate this requirement, for each prerequisite, corequisite, or advisory designated, identify at least three Assignments, Objectives, SLOs, and/or other skills that students will learn in the prerequisite/corequisite/advisory that is essential to the successful completion of the proposed course, and align it with the Assignment/Objective/SLOs/other skill in the proposed course that builds upon that knowledge.

Query output

Cuesta General Education

General Education Categories (effective 2025)

To propose adding a course to one of the eight G.E. Areas, please review the Area Rubric definitions, objectives, and/or core competencies that the course may satisfy. Click on Help link for documents for individual G.E. Areas to complete and upload to the Attached Files section of this proposal.

General Education Categories and Requirements (ended 2025)

General Education Learning Outcomes

Please select the General Education Learning Outcome(s) that this course supports:

Through this program, students will acquire the ability to :

Graduation Requirements (ended 2025)

Transfer General Education

General Education

Distance Education

Modality Limit

See Current Caps Policy for details of proposing different DE Modality Limits than Face-to-face Modality Limits: [AP 4023 Establishment and Modification of Course Caps](#).

Modality specific terms will be used when a course has been approved in Distance Education mode. Since a face-to-face (f2f) modality (even excluding enrollment maximum differences because of room space) may be lower than the Course Cap (if the DE modality is higher) and a face-to-face modality may not be an Enrollment Maximum since these are dependent on classrooms, it is possible to have a f2f number that is neither a course cap nor enrollment max. Therefore, "Face-to-Face Modality Limit" and "DE Modality Limit" will be the terms used when distinguishing between modality limits. Once the course has been approved in DE mode the CurricUNET will automatically add the following categories on the COR to be populated after course cap approvals.

Course Cap 25

Face to face 25

Face to face override

DE modality 25

DE modality override

Distance Education Course Definition

Check the box(es) that apply for the appropriate definition for this course.

- Blended (BL): A combination of face-to-face and distance education with less than 51% of the hours of instruction provided through distance education.

Accessibility

Both the Faculty designing or revising a DE course and the District have an obligation to ensure all course materials have "built in" accommodation for students with disabilities-- i.e. closed or open captioning, descriptive narration and/or interface design/content layout, which is accessible to "industry standard" assistive computer technology in common use by persons with disabilities. (Section 55200) Describe how the DE course design accommodates students with disabilities. Web pages must conform to the standards for accessibility as specified in Section 508 of the Rehabilitation Act (<http://www.access-board.gov/sec508/guide/1194.22.htm>). There are several ways to check web pages for compliance with these standards. Cuesta College policies and resources are available at <http://academic.cuesta.edu/acasupp/DSPS/curbcut/index.htm>.

Describe how the DE course design and any built-in accommodation to course materials accommodate students with disabilities; see Curriculum Handbook for Title V requirements

Use Suggested Language Yes

Create My Own Response No

Enter own language/description

Check the boxes of the types of accessibility that will be used in this course

- Content within the shell is accessible
- Pdf files are accessible
- Videos will be captioned
- Handouts will be formatted for a screen reader
- Images will have ALT tags and descriptions

Regular and Effective Contact

Any portion of a course conducted through distance education shall include regular effective contact between instructor and students, through group or individual meetings, orientation and review sessions, supplemental seminar or study sessions, field trips, library workshops, telephone contact, correspondence, voice mail, e-mail, or other activities (55204-a). Describe how regular and effective contact among instructor and students will be facilitated. Effective contact between instructor and students is based on timeliness of delivery (response to an emailed student question is usually within two working days), accuracy of the subject matter, and provision in a manner and medium appropriate to the significance of the message and the abilities of the student. Describe specific synchronous and asynchronous modes.

Use Suggested Language Yes

Create My Own Response No

Enter own language/description

Instructor-to-student Contact

- Online threaded discussion forums with appropriate instructor feedback
- Daily No
- Weekly Yes
- Bi-weekly No

Other No

Description of Other Instructor-to-Student Contact

- Email

Daily No

Weekly Yes

Bi-weekly No

Other No

Description of Other Instructor-to-Student Contact

- Messaging to individual students

Daily No

Weekly Yes

Bi-weekly No

Other No

Description of Other Instructor-to-Student Contact

- Constructive and timely feedback on student work

Daily No

Weekly Yes

Bi-weekly No

Other No

Description of Other Instructor-to-Student Contact

- Office hours (asynchronous or synchronous)- in person or electronic

Daily No

Weekly Yes

Bi-weekly No

Other No

Description of Other Instructor-to-Student Contact

- Created materials (written, recorded, broadcast, etc.) that, combined with other course materials, creates the "virtual equivalent" of face-to-face class

Daily No

Weekly Yes

Bi-weekly No

Other No

Description of Other Instructor-to-Student Contact

- Facilitating student to student contact

Daily No

Weekly Yes

Bi-weekly No

Other No

Description of Other Instructor-to-Student Contact

Student-to-student Contact

- Email

Daily No

Weekly No

Bi-weekly No

Other No

Description of Other Student-to-Student Contact

- Online threaded discussion forums

Daily No

Weekly Yes

Bi-weekly No

Other No

Description of Other Student-to-Student Contact

Adaptations of Methods, Materials, and Delivery of Instruction

DE Instructors Help Sheet: The DE course design must adapt the learning environment present in a traditional course. Describe both the adaptation of the method of instruction and the communication and computing technologies used to deliver the instruction. If the delivery system is not an industry standard, such as Moodle, provide some detail in describing its features and their function. Also describe any special software or technology students will have to purchase and if the technology will be provided in any campus labs. Indicate if the text and materials of the traditional class are being supplemented or replaced.

Use Suggested Language Yes

Create My Own Response No

Enter own language/description

Adaptations of Assignments and Activities

Describe how assignments and activities will be disseminated. If the Course Outline of record indicates that the course requires critical thinking, special presentations, or activities, describe how they will be incorporated.

Select how Assignments and Activities will be adapted

- Cuesta College LMS formatting used
- Assignments uploaded into the Cuesta College LMS
- Students will "talk" to each other via discussion boards
- Web-specific links used
- Required text reading
- Class presentations using software

Other No

Other Assignments

Adaptations to Methods of Evaluation

Describe how the instructor will appropriately verify that students have met the course objectives, including if proctored exams will be required and if so, how they will be administered. Specifically describe any special software or process used in the adaptation. Specifically describe any special software or process used in the adaptation.

This might include:

- proctored exams
- timed-online or password-protected testing environments
- comparing submitted papers with sample writing assignments or discussion posts
- use of plagiarism detection software (such as TurnItIn.com)
- face-to-face demonstrations of knowledge or skill

Use Suggested Language Yes

Create My Own Response No

Enter own language/description

Method of Evaluation DE

Select how Methods of Evaluation will be adapted and indicate if plagiarism will be evaluated

- Assignments uploaded or typed into the Cuesta College LMS

Plagiarism Evaluation Yes

- Student interaction via discussion boards

Plagiarism Evaluation Yes

- Timed-online or password-protected testing environments

Plagiarism Evaluation Yes

- Face-to-face demonstrations of knowledge or skill

Plagiarism Evaluation Yes

Other No

Plagiarism Evaluation

Student Authentication

View the student authentication PowerPoint presentation: <http://www.3cmediasolutions.org/privid/9183?key=c8edd9a986ea954be6cf0ea456f3e99bc977737a> Select the method(s) used to authenticate students in this course

Teach course in Cuesta College's current LMS Yes

Require monitored exams with Student ID No

Use plagiarism detection software Yes

Provide clear examples of appropriate and inappropriate behavior regarding academic honesty Yes

Require written acceptance of appropriate academic honesty policies by students Yes

Practice regular and effective instructor-student contact Yes

Provide multiple opportunities for authentic assessment No

Employ multiple assessment strategies to maintain active student engagement Yes

Other No

Description of other

Correspondence Education

Instruction Materials

All materials will be due to the CE Program by the CE deadline. All instructional materials must be in print form except for nonprint requirements specified below. Instructional materials will be reviewed, approved, copied/purchased, and delivered to the correctional facility prior to the start of the semester. All course materials will be ADA compliant.

The correctional facility will review all materials delivered to incarcerated students to ensure compliance with CDCR Title 15 criteria (no communication that does not directly relate to the subject and content of the course, no personal communication, no personal stories, and strictly no mention of topics related to sex, violence, legal matters, law enforcement, which again is not specifically and directly part of the course). All submitted research articles, topical papers, and materials needed for students to complete course requirements will be selected by the instructor with those content requirements in mind.

Incarcerated students may not have access to computers and internet resources. All readings for research projects will be supplied in the instructional materials packet. This may include but is not limited to passages from the textbook, journal articles, or other printed materials. Expectations for students to follow a specific writing format style (MLA, APA, Chicago, etc.) will be clearly communicated on the assignment or syllabus.

Course materials and assignments will attempt to replicate the engagement and thought processes that occur during instructor-led discussions in the face-to-face and distance education modalities.

Materials that can be delivered to your students for correspondence instruction include:

- **Course Syllabus**- The syllabus serves as the comprehensive guide for the student to complete the course independently. It shall provide clear indications of all course requirements, including but not limited to:
 - the CE communication process (including Correspondence Form)
 - guidance for course pacing and workflow
 - required readings
 - assignment expectations
- **Course Textbook/Required Readings** — The textbook name, edition, and availability will be confirmed by the institution's CE program staff.
- **Print Handout Materials** — Typical supplemental handout materials may include, but are not limited to:
 - Slide presentations (printed)
 - Instructional resources from the textbook publisher — downloaded from an authorized website.
 - Research articles related to the course — downloaded from an authorized website

Optional:

- **Nonprint Materials** — All nonprint materials will be ADA compliant (e.g. videos will be close-captioned).

Optional:

Assignments

A description of each assignment for the entire term will be clearly indicated on the syllabus and the instructional materials packet. Students may be required to sign an honor statement.

Course Quiz/Exam/Assessment Proctoring Requirements (select one option)

Regular Effective Contact

Education Code requires that correspondence education ensures substantive faculty-student interaction.

Per proposed Title 5 regulations: *"Both modalities [Distance Education and Correspondence] require faculty to play a proactive role in the development and progress of learning through substantive feedback and the development of responsive assignments and course materials...."*

Instructor communication will attempt to replicate the engagement and thought processes that occur during instructor-led discussions in the face-to-face and distance education modalities. Substantive faculty-student interaction is provided through:

- the syllabus
- course instructional materials
- assignments

- quizzes/exams/assessments
- constructive and timely feedback on student work
- Correspondence Form (individual or to the class)

Substantive faculty-student interaction is defined as a minimum of nine (9) different contacts using the elements above.

Optional

Materials Fee

is a material fee recommended for this course? No

Attached Files

Please visit the Cuesta CurrlQunet Meta Help Page for help, instructions, and tutorials:

https://www.cuesta.edu/meta/docs/Help_Documents.pdf

Attached File

ASSIST

Please visit the Cuesta CurrlQunet Meta Help Page for help, instructions, and tutorials:

https://www.cuesta.edu/meta/docs/Help_Documents.pdf

Queue for Assist No

Last Request From Queue 10/29/2025

Last Direct Request

ASSIST Preview

Please visit the Cuesta CurrlQunet Meta Help Page for help, instructions, and tutorials:

https://www.cuesta.edu/meta/docs/Help_Documents.pdf

Prefix AERO

Course Number 100

Content

1. Maintenance and Preflight Inspection Procedures

- Regulatory Requirements for Maintenance
- Preflight Inspection
- Postflight Inspection
- Manufacturer Inspection Intervals
- Battery Maintenance
- Flight and Maintenance Logs
- Maintenance, Repair, and Replacement of Components
(Obj 1, 5)

2. UAV Flight Operations

- Takeoff
- Aircraft Control
- Preflight Programming
- Turns around a point
- Aircraft Positioning
- Altitude Control
- Sensor Zoom/Pan/Tilt
- Airspeed Control
- Approach
- Landing
(Obj 5)

3. Emergency Procedures

- Contingency Planning
- Ground Emergencies
- Inflight Emergencies
- Emergency Communications
(Obj 4)

4. Human Factors

- Health and Environmental Factors
- Decision Making and Judgment
- Risk Management and Aeronautical Decision Making
(Obj 3)

5. Aircraft Performance

- Weight and Balance Calculations
- Density Altitude
- Takeoff and Climb Performance
- Endurance
- Landing Distance
- Aerodynamics
- Ground Effect

(Obj 2)

6. Aviation Weather Theory, Sources, and Effects on Performance

- Fronts, Stability, Clouds, and Thunderstorms
- Fog
- Structural Icing and Frost
- Weather Sources for Aviation Operations
- The AWC Home Page (NWS Aviation Weather Center)
- Practical Sources for the Remote Pilot
- Automated Weather Reporting Stations

(Obj 2, 4)

7. Airport and Off-Airport Operations

- Operations at Non-Tower Airports
- Operations at Airports with Control Towers
- Off-Airport Operations
- Flying at Night

(Obj 4)

8. Airspace and Navigation

- Aeronautical Charts
- Knowing Where You Can Fly
- Geographical Coordinates
- Chart Reading
- Airspace
- Correcting for Wind Drift
- Special Use Airspace

(Obj 4)

9. Regulations, Procedures, and Other Aeronautical References

- UAS Terms and Abbreviations
- Types of UAV, Components and Uses of UAV
- Government Regulations and Other Procedures Publications
- FAA Chart Supplements
- Advisory Circulars
- Notices to Airmen
- Aviation Media

(Obj 4)

Lab Content

Lab content embedded in course content.

Assignments

Critical Thinking Assignments (minimum 2 examples):

Perform preflight calculations to determine if flight can be safely conducted based on aircraft performance, knowledge of aviation weather, forecast products, and airspace.

(Obj 2, 3, 4)

Independent Work (minimum 2 examples):

Perform small unmanned aerial vehicle flight to the standard test methods found in NIST sUAS open lane Level II standards by maneuvering a sUAS over selected targets in specific sequence and land aircraft safely.

(Obj 5)

Critical Thinking Assignments (minimum 2 examples):

Based on knowledge of the weather, approved aviation weather forecast sources, and mission requirements, determine if flight can be safely conducted near an airport with an operating control tower.

(Obj 4)

Independent Work (minimum 2 examples):

Perform a technical preflight inspection of aircraft in accordance with the manufacturer's recommendations, determine if safe for flight, and calculate expected in-flight endurance.

(Obj 1, 2, 3)

Course Description

Prepares students to meet the Federal Aviation Administration requirements for certification to operate Unmanned Aircraft Vehicles under 55 pounds for commercial operations pursuant to 14 CFR 107. Emphasizes the safe, legal, and ethical aspects of UAV operations in civil, public, and commercial settings. Includes special emphasis on capabilities and limitations of optical payloads and sensors, flight operation planning, and safety. Prepares students to safely operate UAV through all phases of flight to National Institute of Standards and Technology Level 2 Open Area Maneuvering remote pilot proficiency standards.

Lecture Hours

36.00

Lab Hours

54.00

Outline Approval Date

Outline Effective Date

Prerequisites

None

Corequisites

None

Advisories

None

Objectives

1. Perform a flight of an unmanned aerial vehicle to the "Open Lane Level II NIST Standard Test Methods for small Unmanned Aircraft Systems (sUAS). The U.S. National Institute of Standards and Technology (NIST) quantitatively evaluates various system capabilities and remote pilot proficiency.
2. Demonstrate knowledge of regulations, airport operations procedures, aviation weather, radio communication procedures, emergency procedures, airspace, and navigation to FAA Part 107 standards.
3. Implement appropriate decision making and judgment to mitigate risk based on human factor considerations.
4. Calculate the performance of UAV to include weight and balance, takeoff and climb, endurance, and aerodynamic effects based on actual and forecast aviation weather.
5. Perform preflight inspection to identify maintenance deficiencies that adversely affect aircraft performance.

Instruction Methods

Lab, Lecture, Distance Education, DE Synchronous, DE Asynchronous, Lecture/Lab

Evaluation Methods

- Performance Exams
- Quizzes/Exams
- Field Work
- Essay Exams
- Class Performance(s)

Other Texts**Textbooks:**

David Ison *The Complete Remote Pilot* (3rd/e). Aviation Supplies & Academies, Inc., Newcastle, Washington, (2025).

Manuals:

Mavic Drone Manua, DJI, DJI, 2025

Other:

Access to online aviation weather services provided by the National Weather Service - Aviation Weather Center.